Special Functions

S.1

Solution:

Using the definition of $\Gamma(z)$:

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} \, \mathrm{d}t$$

it can be shown that $\Gamma(z+1) = z\Gamma(z)$:

$$\Gamma(z+1) = \int_0^\infty t^z e^{-t} \, \mathrm{d}t = \underbrace{\left[-t^z e^{-t}\right]_0^\infty}_{=0} + \int_0^\infty z t^{z-1} e^{-t} \, \mathrm{d}t =$$

$$= z \int_0^\infty t^{z-1} e^{-t} \, \mathrm{d}t = z\Gamma(z)$$

We also have

$$\Gamma(1) = \int_0^\infty e^{-t} \, \mathrm{d}t = \left[-e^{-t}\right]_0^\infty = 1$$

This gives

$$\Gamma(3) = \Gamma(2+1) = 2\Gamma(2) = 2\Gamma(1+1) = 2 \cdot 1 \cdot \Gamma(1) \rightarrow \boxed{\Gamma(3) = 2}$$

Next

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-\frac{1}{2}} e^{-t} \, \mathrm{d}t \rightarrow \left[t = x^2 \rightarrow \mathrm{d}t = 2x \, \mathrm{d}x\right] = 2 \int_0^\infty x x^{-1} e^{-x^2} \, \mathrm{d}x =$$

$$= 2 \int_0^\infty e^{-x^2} \, \mathrm{d}x = 2\frac{\sqrt{\pi}}{2} \rightarrow \boxed{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$$

Finally we will use the definition for $\Gamma(z), \mathrm{Re}\ z \leq 0$:

$$\Gamma(z) = \frac{\Gamma(z+n)}{(z+n-1)...(z+1)z}, \quad n \text{ chosen such that } \mathrm{Re}(z+n) > 0$$

This gives

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(-\frac{1}{2}+1\right)}{-\frac{1}{2}} = \frac{\Gamma\left(\frac{1}{2}\right)}{-\frac{1}{2}} \rightarrow \boxed{\Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi}}$$

□

S.5

Solve the Abelian integral

$$\int_0^x \frac{u(y)}{\sqrt{x-y}}\,\mathrm{d}y = 1, \quad x > 0$$

Solution:

$$\frac{\Gamma\left(\frac{1}{2}\right)}{s^{\frac{1}{2}}} U(s) = F(s)$$

$$\downarrow$$

$$\frac{\Gamma\left(\frac{1}{2}\right)}{s^{\frac{1}{2}}} U(s) = \frac{1}{s}$$

$$\downarrow$$

$$U(s) = \frac{1}{\Gamma\left(\frac{1}{2}\right)s^{\frac{1}{2}}} = \frac{1}{\Gamma\left(\frac{1}{2}\right)^2} \frac{\Gamma\left(\frac{1}{2}\right)}{s^{\frac{1}{2}}} \xrightarrow{\mathcal{L}^{-1}} u(x)\theta(x) = \frac{1}{\pi} x^{-\frac{1}{2}} \theta(x)$$

$$\downarrow$$

$$\boxed{u(x) = \frac{1}{\pi\sqrt{x}}, \quad x > 0}$$

□

S.6

Solution:

□

S.7

Show that

$$\sum_{-\infty}^{\infty} |J_n(r)|^2 = 1$$

Solution:

Using the Fourier expansion

$$e^{ir\sin\theta} = \sum_{-\infty}^{\infty} J_n(r) e^{in\theta}$$

and Parseval's Theorem

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |e^{ir\sin\theta}|^2 \,\mathrm{d}\theta = \sum_{-\infty}^{\infty} |J_n(r)|^2$$

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \overline{e^{ir\sin\theta}} e^{ir\sin\theta} \,\mathrm{d}\theta = \frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{d}\theta = \frac{1}{2\pi} 2\pi = 1$$

$$\downarrow$$

$$\boxed{\sum_{-\infty}^{\infty} |J_n(r)|^2 = 1}$$

□

S.11

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{J_\nu(x)}{x^\nu}\right) = -\frac{J_{\nu+1}(x)}{x^\nu}$$

(b) Use (a) and

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

to calculate $J_{\frac{3}{2}}(x)$

(c) Show that $J_0'(x) = -J_1(x)$

Solution:

(a) The power series is

$$J_\nu(x) = \left(\frac{x}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k!\Gamma(\nu+k+1)}\left(-\frac{x^2}{4}\right)^k$$

$$\downarrow$$

$$\frac{J_\nu(x)}{x^\nu} = \left(\frac{1}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k!\Gamma(\nu+k+1)}\left(-\frac{x^2}{4}\right)^k$$

This can be differentiated termwise as

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{J_\nu(x)}{x^\nu}\right) = \left(\frac{1}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{-2xk}{4k!\Gamma(\nu+k+1)}\left(-\frac{x^2}{4}\right)^{k-1} =$$

$$= -\frac{x}{2}\left(\frac{1}{2}\right)^\nu \underbrace{\sum_{k=1}^{\infty}}_{*} \frac{1}{(k-1)!\Gamma(\nu+k+1)}\left(-\frac{x^2}{4}\right)^{k-1} =$$

$$= -\frac{1}{x^\nu}\left(\frac{x}{2}\right)^{\nu+1} \sum_{k=1}^{\infty} \frac{1}{(k-1)!\Gamma(\nu+k+1)}\left(-\frac{x^2}{4}\right)^{k-1} \to [k \to k+1] \to$$

$$\to -\frac{1}{x^\nu}\left(\frac{x}{2}\right)^{\nu+1} \sum_{k=0}^{\infty} \frac{1}{k!\Gamma(\nu+1+k+1)}\left(-\frac{x^2}{4}\right)^{k} = \boxed{-\frac{J_{\nu+1}(x)}{x^\nu}}$$

* (The k=0 contribution is zero, since it is a constant and was differentiated.)

(b)Setting $\nu = \frac{1}{2}$ we get

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}}\right) = -\frac{J_{\frac{3}{2}}(x)}{x^{\frac{1}{2}}}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\sqrt{\frac{2}{\pi x}}\sin x}{x^{\frac{1}{2}}}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{\frac{2}{\pi}}\frac{\sin x}{x}\right) = \sqrt{\frac{2}{\pi}}\left(\frac{x\cos(x) - \sin(x)}{x^2}\right)$$

$$\downarrow$$

$$J_{\frac{3}{2}}(x) = -x^{\frac{1}{2}}\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}}\right) = x^{\frac{1}{2}}\sqrt{\frac{2}{\pi}}\left(\frac{\sin(x) - x\cos(x)}{x^2}\right) = \sqrt{\frac{2x}{\pi}}\left(\frac{\sin x - x\cos x}{x^2}\right)$$

(c) Using the result from (a)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{J_0(x)}{x^0}\right) = -\frac{J_1(x)}{x^0} \to \boxed{J_0'(x) = -J_1(x)}$$

□

S.12

Solution:

□

S.13

(a) $r^2y'' + ry' = 0$

(b) $r^2y'' + ry' - y = 0$

(c) $r^2y'' + ry' + r^2y = 0$

Solution:

Using a definition that the general solution to Bessel's differential equations

$$u'' + \frac{1}{r}u' + \left(\lambda - \frac{\nu^2}{r^2}\right)u = 0$$

are

$$\begin{aligned} &aJ_\nu\left(\sqrt{\lambda}r\right) + bY_\nu\left(\sqrt{\lambda}r\right), && \text{if } \lambda > 0 \\ &ar^\nu + br^{-\nu}, && \text{if } \lambda = 0, \nu \neq 0 \\ &a + b\ln r, && \text{if } \lambda = \nu = 0 \end{aligned}$$

(a)

$$r^2y'' + ry' = 0 \to y'' + \frac{1}{r}y' = 0, \quad \lambda = \nu = 0$$

$$\boxed{y(r) = a + b\ln r}$$

(b)

$$r^2y'' + ry' - y = 0 \to y'' + \frac{1}{r}y' - \frac{1}{r^2}y = 0, \quad \lambda = 0, \nu = 1$$

$$\boxed{y(r) = ar + \frac{b}{r}}$$

(c)

$$r^2y'' + ry' + r^2y = 0 \to y'' + \frac{1}{r}y' + y = 0, \qquad \lambda = 1, \nu = 0$$

$$\boxed{y(r) = aJ_0(r) + bY_0(r)}$$

□

S.14

$$\mathcal{A}u = -\frac{1}{x}(xu')'$$

$$D_{\mathcal{A}} = \{u \in \mathcal{C}^2(I) \mid u \text{ bounded near } 0, u'(2) = 0\}$$

(a) Determine a scalar product in which $\mathcal{A}$ is symmetric and positive semidefinite

(b) Determine the eigenvalues and eigenvectors of $\mathcal{A}$.

(c) Determine numerically the three lowest eigenvalues.

(d) Let $u(x) = x$. Write down u's expansion in eigenfunctions to $\mathcal{A}$. The coefficients should be given in integral form.

Solution:

(a)

$$(u \mid \mathcal{A}v) = \int_0^2 \overline{u(x)}\mathcal{A}v(x)w(x)\,\mathrm{d}x$$

with $w(x) = x$

$$(u \mid \mathcal{A}v) = -\int_0^2 \overline{u}(xv')'\,\mathrm{d}x = -\underbrace{[\overline{u}(xv')]_0^2}_{=0} + \int_0^2 (\overline{u'}x)v'\,\mathrm{d}x =$$

$$= \underbrace{\int_0^2 (x\overline{u'})v'\,\mathrm{d}x}_{=(*)} = \underbrace{\left[(x\overline{u'})v\right]_0^2}_{=0} - \int_0^2 \left(x\overline{u'}\right)' v\,\mathrm{d}x = \boxed{(\mathcal{A}u \mid v)}$$

To show that it's positive semidefinite, start from (*) and set $u = v$:

$$\int_0^2 x\overline{u'}u'\,\mathrm{d}x = \int_0^2 x|u'|^2\,\mathrm{d}x \geq 0$$

(b) Setting

$$\mathcal{A}u = \lambda u \to -\frac{1}{x}(xu')' = -\frac{1}{x}(u' + xu'') = \lambda u$$
$$\downarrow$$
$$-\frac{1}{x}xu'' - \frac{1}{x}u' - \lambda u = 0 \to u'' + \frac{1}{x}u' + \lambda u = 0$$

which has the solutions (according to the definition of the Bessel's differential equation solutions):

$$aJ_0\left(\sqrt{\lambda}x\right) + bY_0\left(\sqrt{\lambda}x\right), \qquad \lambda > 0, \nu = 0$$

Since Y_0 is unbounded near $x = 0$, b has to be zero to fulfill the requirement of u being bounded near zero.

$$u(x) = aJ_0\left(\sqrt{\lambda}x\right)$$
$$u'(2) = \sqrt{\lambda}aJ_0'\left(\sqrt{\lambda}2\right) = 0$$

This gives $\lambda_0 = 0$. The rest of the eigenvalues are given by

$$J_0'\left(2\sqrt{\lambda_k}\right) = 0 \to \lambda_k = \left(\frac{\beta_{0,k}}{2}\right)^2$$

The eigenfunctions are then given by

$$\varphi_{k(x)} = J_0\left(\frac{\beta_{0,k}}{2}x\right)$$

(c)

$$\lambda_0 = 0$$
$$\lambda_1 = \left(\frac{3.8317}{2}\right)^2 = 3.67$$
$$\lambda_2 = \left(\frac{7.0156}{2}\right)^2 = 12.31$$

□

S.15

Determine a non-trivial solution to the eigenvalue problem

$$-\Delta u = \lambda u, \quad \text{in } \Omega$$
$$u = 0, \qquad \text{on } \partial\Omega$$

where Ω is the circle sector $0 < r < 1, 0 < \theta < \frac{\pi}{3}$

Solution:

We have the conditions for $u(r,\theta)$

$$\begin{aligned} u(r,0) &= u\left(r,\frac{\pi}{3}\right) = 0 \\ u(1,\theta) &= 0 \end{aligned}$$

The Laplacian in cylindrical coordinates is

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}$$

Expressing $u(r,\theta)$ with separation of variables gives

$$u(r,\theta) = R(r)\Theta(\theta)$$

Putting this into the Laplacian gives

$$R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta'' = -\lambda R\Theta$$
$$\downarrow$$
$$\frac{R''}{R} + \frac{1}{r}\frac{R'}{R} + \frac{1}{r^2}\frac{\Theta''}{\Theta} = -\lambda$$
$$\downarrow$$
$$r^2\frac{R''}{R} + r\frac{R'}{R} + \lambda r^2 = -\frac{\Theta''}{\Theta}$$
$$\downarrow$$
$$\frac{r^2}{R}\left(R'' + \frac{1}{r}R' + \lambda R\right) = -\frac{\Theta''}{\Theta}$$
$$\downarrow$$
$$\frac{r^2}{R}\left(\frac{1}{r}(rR')' + \lambda R\right) = -\frac{\Theta''}{\Theta}$$

LHS depend only on r while RHS depend only on θ, therefore both must be a constant

$$\frac{r^2}{R}\left(\frac{1}{r}(rR')' + \lambda R\right) = \beta \to \frac{1}{r}(rR')' + \lambda R = \frac{\beta R}{r^2}$$
$$-\frac{\Theta''}{\Theta} = \beta$$
$$\downarrow$$
$$\begin{cases} \Theta'' = -\beta\Theta \\ R'' + \frac{1}{r}R' + R\left(\lambda - \frac{\beta}{r^2}\right) = 0 \end{cases}$$

Solution to Θ:

$$\Theta(\theta) = A\cos\left(\sqrt{\beta}\theta\right) + B\sin\left(\sqrt{\beta}\theta\right)$$
$$\Theta(0) = A \rightarrow A = 0$$
$$\Theta\left(\frac{\pi}{3}\right) = B\sin\left(\sqrt{\beta}\frac{\pi}{3}\right) = 0 \rightarrow \sqrt{\beta}\frac{\pi}{3} = k\pi \rightarrow \beta = 9k^2, \quad k = 1, 2, 3...$$
$$\boxed{\Theta(\theta) = \sin\left(\sqrt{\beta}\theta\right)}$$

Put the β into the equation for $R(r)$ and we get Bessel's differential equation:

$$R'' + \frac{1}{r}R' + \left(\lambda - \frac{9k^2}{r^2}\right)R = 0, \nu^2 = 9k^2 \rightarrow \nu = 3k$$

This has the solution

$$R(r) = a_k J_{3k}\left(\sqrt{\lambda}r\right) + b_k Y_{3k}\left(\sqrt{\lambda}r\right)$$

Since $R(r)$ has to be bounded, $b_k = 0$

$$R(r) = a_k J_{3k}\left(\sqrt{\lambda_k}r\right)$$
$$R(1) = 0 \rightarrow \lambda_k = \alpha_{3k,n}^2$$

The full solution is then

$$\boxed{\begin{aligned} u(r,\theta) &= a_k J_{3k}\left(a_{3k,n}r\right)\sin(3k\theta) \\ \lambda_{kn} &= a_{3k,n}^2, \quad k,n = 1,2,3... \end{aligned}}$$

□

S.16

The spherical Bessel functions are defined as

$$j_l(x) = \sqrt{\frac{\pi}{2x}} J_{l+\frac{1}{2}}(x)$$

Calculate j_0, j_1, and j_2 by using

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}}\sin(x)$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{J_\nu(x)}{x^\nu}\right) = -\frac{J_{\nu+1}(x)}{x^\nu}, \quad \nu = 0, 1, 2...$$

Solution:

$$\boxed{j_0(x) = \sqrt{\frac{\pi}{2x}} J_{\frac{1}{2}}(x) = \sin(x)}$$

$$\boxed{j_1(x) = \sqrt{\frac{\pi}{2x}} J_{\frac{3}{2}}(x) = \frac{\sin(x) - x\cos(x)}{x^2}}$$

$$j_2(x) = \sqrt{\frac{\pi}{2x}} J_{\frac{5}{2}}(x) = \sqrt{\frac{\pi}{2x}} \left(-x^{\frac{3}{2}}\right) \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{J_{\frac{3}{2}}(x)}{x^{\frac{3}{2}}} \right)$$

$$\frac{J_{\frac{3}{2}}(x)}{x^{\frac{3}{2}}} = \frac{1}{x^{\frac{3}{2}}} \sqrt{\frac{2x}{\pi}} \left(\frac{\sin(x) - x\cos(x)}{x^2} \right) = \sqrt{\frac{2}{\pi}} \left(\frac{\sin(x) - x\cos(x)}{x^3} \right)$$

$$\frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{J_{\frac{3}{2}}(x)}{x^{\frac{3}{2}}} \right) = \sqrt{\frac{2}{\pi}} \left(\frac{x^4 \sin(x) - 3x^2 \sin(x) + 3x^3 \cos(x)}{x^6} \right)$$

$$\boxed{j_2(x) = \left(\frac{(3 - x^2)\sin(x) - 3x\cos(x)}{x^3} \right)}$$

□

S.18

Determine the general solution to

(a) $r^2 y'' + 2ry' = 0$

(b) $r^2 y'' + 2ry' - 2y = 0$

(c) $r^2 y'' + 2ry' + r^2 y = 0$

Solution:

The general solution to the operator

$$\mathcal{A} = y'' + \frac{2}{r} y' + \left(\lambda - \frac{l(l + \frac{1}{2})}{r^2} \right) y$$

is

$$\begin{cases} aj_l\left(\sqrt{\lambda}r\right) + by_l\left(\sqrt{\lambda}r\right), & \lambda > 0 \\ ar^l + br^{-l-1}, & \lambda = 0,\, l \neq -\frac{1}{2} \\ \frac{a+b\ln(r)}{\sqrt{r}}\,, & \lambda = 0,\, l = -\frac{1}{2} \end{cases}$$

(a)

$$r^2y'' + 2ry' = 0 \to y'' + \frac{2}{r}y' = 0$$

Here we have $\lambda = 0, l = 0$, so the solution is:

$$y(r) = a + \frac{b}{r}$$

(b)

$$r^2y'' + 2ry' - 2y = 0 \to y'' + \frac{2}{r}y' - \frac{2}{r^2}y = 0$$
$$\downarrow$$
$$y'' + \frac{2}{r}y' + \left(0 - \frac{2}{r^2}\right)y = 0$$

Here we have $\lambda = 0, l(l+1) = 2 \to l = 1$:

$$y(r) = ar + \frac{b}{r^2}$$

(c)

$$r^2y'' + 2ry' + r^2y = 0 \to y'' + \frac{2}{r}y' + y = 0$$
$$\downarrow$$
$$y'' + \frac{2}{r}y' + \left(1 - \frac{0\cdot(0+1)}{r^2}\right)y = 0$$

Here we have $\lambda > 0$ and $l = 0$ (or $l = -1$)

$$y(r) = aj_0(r) + by_0(r)$$

□

S.20

Let $\Omega =$ unit sphere in $\mathbb{R}^3$,

$$\mathcal{A} = -\Delta$$
$$D_{\mathcal{A}} = \{u \in C^2(\Omega) \mid u = 0 \text{ on } \partial\Omega\}.$$

(a) Determine the radial (θ, ϕ independent) eigenfunctions and eigenvalues.

(b) Expand the function $1(x, y, z) \equiv 1$ using eigenfunctions to $\boldsymbol{\mathcal{A}}$.

Solution:

The radial component of the Laplacian in spherical coordinates is

$$\Delta_r = u'' + \frac{2}{r}u'$$

So we want to solve

$$\boldsymbol{\mathcal{A}}u = \lambda u \to -\Delta_r u = \lambda u$$
$$\downarrow$$
$$u'' + \frac{2}{r}u' + \lambda u = 0$$
$$\downarrow$$
$$u'' + \frac{2}{r}u' + \left(\lambda - \frac{l(l+1)}{r^2}\right)u = 0$$

Here we have $\lambda > 0$ and $l = 0$ and we use the general solution to this and get

$$u(r) = aj_0\left(\sqrt{\lambda}r\right) + by_0\left(\sqrt{\lambda}r\right)$$

Since $u(r)$ has to be bounded at $r = 0$, the $y_0\left(\sqrt{\lambda}r\right)$-term has to disappear, so $b = 0$ and our final solution is

$$u(r) = aj_0\left(\sqrt{\lambda}r\right)$$

The eigenvalues can be found from

$$u(1) = 0 \to aj_0\left(\sqrt{\lambda}\right) = 0$$

which means $\sqrt{\lambda}$ has to make $j_0 = 0$. But $j_0\left(\sqrt{\lambda}r\right) = \frac{\sin\left(\sqrt{\lambda}r\right)}{r}$ so that gives

$$\sqrt{\lambda} = k\pi$$
$$\downarrow$$
$$\boxed{\lambda_k = k^2\pi^2}$$

and the eigenfunctions are

$$\boxed{\varphi_k(r) = \frac{\sin(k\pi r)}{r}}$$

(b)

$$1 = \sum_{k=1}^{\infty} c_k \varphi_k(r)$$

$$c_k = \frac{(\varphi_k \mid 1)}{(\varphi_k \mid \varphi_k)} = \frac{\int_0^1 r\sin(k\pi r)\,\mathrm{d}r}{\int_0^1 \sin^2(k\pi r)\,\mathrm{d}r} = \frac{\left[\frac{-r\cos(k\pi r)}{k\pi}\right]_0^1 + \underbrace{\int_0^1 \frac{\cos(k\pi r)}{k\pi}\,\mathrm{d}r}_{=0}}{\int_0^1 \left(\frac{1}{2} - \underbrace{\frac{\cos(2k\pi r)}{2}}_{\int = 0}\right)\mathrm{d}r} =$$

$$= -2\frac{\cos(k\pi)}{k\pi} \to \boxed{c_k = \frac{2}{k\pi}(-1)^{k+1}}$$

□